

User Interface Design Documentation

Project Name: Phoneword Generator

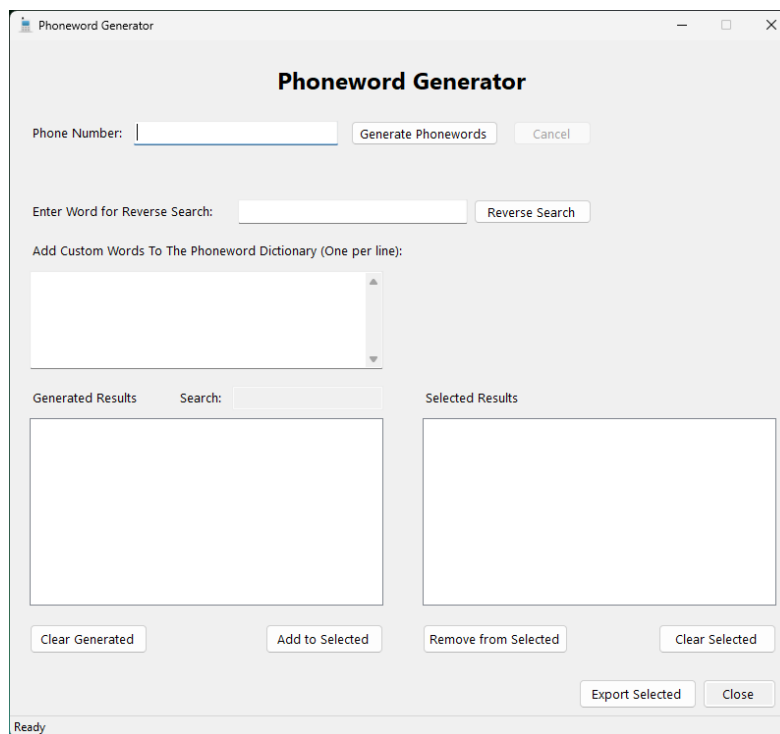
Due Date: February 12, 2025

1. Introduction

The Phoneword Generator helps users turn phone numbers into memorable words (phonewords) and vice versa. This document describes and documents the user interface (UI).

2. Main Screen Overview

The main screen is designed to be easy to use, with all features accessible from a single window.



3. UI Components

A. Title

- **Title Label** (`lblTitle`): Displays "Phoneword Generator" at the top.

B. Phone Number Input

- **Phone Number Label** (`lblPhoneNumber`): Prompts the user to enter a phone number.
- **Phone Number Textbox** (`txtPhoneNumber`): Where the user types the phone number.
 - **Accepts:** Only digits (0-9).
- **Generate Button** (`btnGenerate`): Starts generating phonewords.
- **Cancel Button** (`btnCancel`): Stops the generation process.
 - **Enabled only while generating.**

C. Reverse Search

- **Reverse Search Label** (`lblReverseSearch`): Prompts the user to enter a word.
- **Word Input Textbox** (`txtWordInput`): Where the user types a word to find its phone number.
 - **Accepts:** Only uppercase letters (A-Z).
- **Reverse Search Button** (`btnReverseSearch`): Converts the word to a phone number.

D. Custom Words

- **Custom Words Label** (`lblCustomWords`): Indicates where to add custom words.
- **Custom Words Textbox** (`txtCustomWords`): Users can type their own words to include in generation.
 - **Accepts:** Uppercase letters and new lines.

E. Generated Results

- **Generated Results Label** (`lblGeneratedResults`): Label for the generated phonewords list.
- **Generated List** (`lstGenerated`): Shows phonewords generated from the phone number.
- **Search Generated Label** (`lblSearchGenerated`): Indicates the search box for generated phonewords.
- **Search Generated Textbox** (`txtSearchGenerated`): Lets users search within generated phonewords.
- **Clear Generated Button** (`btnClearGenerated`): Clears the list of generated phonewords.

F. Selected Results

- **Selected Results Label** (`lblSelectedResults`): Label for selected phonewords.
- **Selected List** (`lstSelected`): Shows phonewords the user has selected.
- **Add Button** (`btnAdd`): Adds selected generated phonewords to the selected list.
- **Remove Button** (`btnRemove`): Removes selected phonewords from the selected list.
- **Clear Selected Button** (`btnClearSelected`): Clears all selected phonewords.
- **Export Button** (`btnExport`): Saves selected phonewords to a text file.

G. Status and Progress

- **Status Strip** (`statusStrip`) & **Status Label** (`statusLabel`): Shows messages like "Generating..." or "Completed."
- **Progress Bar** (`progressBar`): Displays progress during generation.

H. Close Application

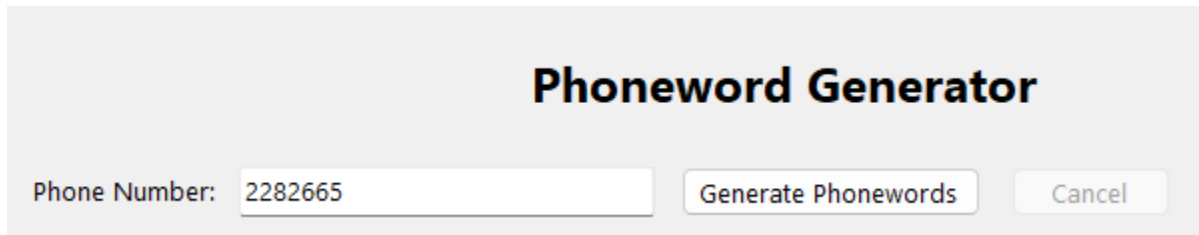
- **Close Button** (`btnClose`): Closes the application.
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4. How to Use the Application

4.1 Generating Phonewords

4.1.1 Enter a Phone Number

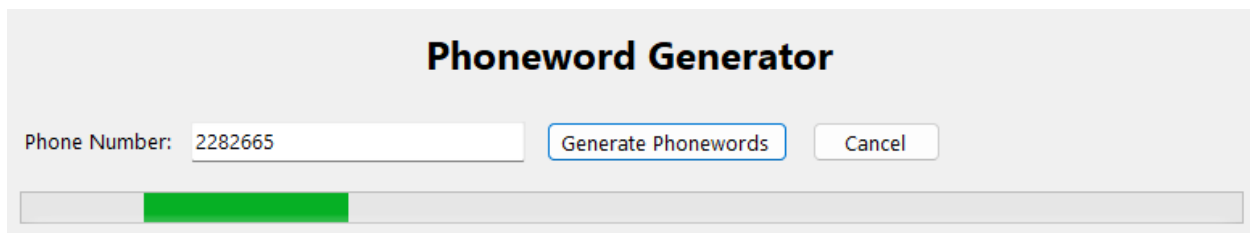
- Type a phone number into the **Phone Number Textbox** (`txtPhoneNumber`).
 - Only numbers (0-9) are allowed.
 - Letters and special characters cannot be entered.



The screenshot shows a window titled "Phoneword Generator". Inside, there is a label "Phone Number:" followed by a text input field containing the number "2282665". To the right of the input field are two buttons: "Generate Phonewords" and "Cancel".

4.1.2 Generate Phonewords

- Click the **Generate Button** (`btnGenerate`) to start generating phonewords.
- The **Progress Bar** appears, showing the generation is in progress.
- The **Cancel Button** (`btnCancel`) becomes active; click it if you need to stop the process.



This screenshot shows the same "Phoneword Generator" window as the previous one, but with a progress bar at the bottom. The progress bar is a horizontal bar with a green segment on the left, indicating progress. The "Generate Phonewords" button now has a blue border, and the "Cancel" button is also active.

4.1.3 View Results

- Once generation is complete:
 - The **Progress Bar** disappears.
 - The **Cancel Button** is disabled.
- Generated phonewords appear in the **Generated List** (`lstGenerated`).
- If there are many results, use the **Search Generated Textbox** to find specific phonewords.

The interface consists of two main panels: "Generated Results" on the left and "Selected Results" on the right. Above the "Generated Results" panel is a "Search:" label followed by a text input field. The "Generated Results" panel contains a scrollable list of phonewords: ACT-AMOK, ACT-BOOK, ACT-CONK, ACT-COOK, ACT-COOL, BAT-AMOK, BAT-BOOK, BAT-CONK, BAT-COOK, BAT-COOL, and CAT-AMOK. Below this list are four buttons: "Clear Generated", "Add to Selected", "Remove from Selected", and "Clear Selected". The "Selected Results" panel is currently empty.

4.2 Managing Generated Phonewords

4.2.1 Search Within Generated Results

- Type letters or words into the **Search Generated Textbox** (`txtSearchGenerated`) to filter the generated phonewords.
 - The list updates automatically to show matching phonewords.

The interface is the same as the previous screenshot, but the "Search:" text input field now contains the word "cool". The "Generated Results" list has been filtered to show only phonewords containing "cool": ABU-COOL, ACT-COOL, BAT-COOL, CAT-COOL, AAT-COOL, AAU-COOL, AAV-COOL, ABT-COOL, ABV-COOL, ACU-COOL, and ACV-COOL. The "Selected Results" panel remains empty, and the buttons at the bottom are the same.

4.2.2 Select Phonewords to Save

- In the **Generated List**, click to select phonewords you like.
 - To select multiple items, hold the **Ctrl** key while clicking each item.
- Click the **Add Button** (`btnAdd`) to move selected phonewords to the **Selected List** (`lstSelected`).

The screenshot displays a user interface for managing phonewords. It features two main panels: 'Generated Results' on the left and 'Selected Results' on the right. Above the 'Generated Results' panel is a search bar labeled 'Search:'. The 'Generated Results' panel contains a list of phonewords: ABU-AMOK, ABU-BOOK, ABU-CONK, ABU-COOK, ABU-COOL, ACT-AMOK, ACT-BOOK, ACT-CONK, ACT-COOK, ACT-COOL, and BAT-AMOK. The 'Selected Results' panel contains a list of phonewords: ACT-COOL, BAT-COOL, CAT-COOL, CAT-BOOK, and CAT-COOK. Below the 'Generated Results' panel are two buttons: 'Clear Generated' and 'Add to Selected'. Below the 'Selected Results' panel are two buttons: 'Remove from Selected' and 'Clear Selected'.

Generated Results	Search:	Selected Results
ABU-AMOK		ACT-COOL
ABU-BOOK		BAT-COOL
ABU-CONK		CAT-COOL
ABU-COOK		CAT-BOOK
ABU-COOL		CAT-COOK
ACT-AMOK		
ACT-BOOK		
ACT-CONK		
ACT-COOK		
ACT-COOL		
BAT-AMOK		

Buttons: Clear Generated, Add to Selected, Remove from Selected, Clear Selected

4.2.3 Clear Generated Results

- Click the **Clear Generated Button** (`btnClearGenerated`) to remove all phonewords from the **Generated List**.
- This also clears any search filters applied.

4.3 Managing Selected Phonewords

4.3.1 Review Selected Phonewords

- The **Selected List** (`lstSelected`) displays the phonewords you've chosen to keep or export.

4.3.2 Remove Phonewords from Selected List

- To remove items:
 - Select the phonewords you want to delete in the **Selected List**.
 - Click the **Remove Button** (`btnRemove`).

The screenshot shows a web interface with two main panels: 'Generated Results' on the left and 'Selected Results' on the right. Above the panels is a search bar labeled 'Search:'. The 'Generated Results' panel contains a list of phonewords: ABU-AMOK, ABU-BOOK, ABU-CONK, ABU-COOK, ABU-COOL, ACT-AMOK, ACT-BOOK, ACT-CONK, ACT-COOK, ACT-COOL, and BAT-AMOK. The 'Selected Results' panel contains a list of phonewords: CAT-COOL, CAT-BOOK, and CAT-COOK. Below the panels are four buttons: 'Clear Generated', 'Add to Selected', 'Remove from Selected' (which is highlighted with a blue border), and 'Clear Selected'.

4.3.3 Clear All Selected Phonewords

- Click the **Clear Selected Button** (`btnClearSelected`) to remove all phonewords from the **Selected List**.

The screenshot shows the same web interface as before, but the 'Selected Results' panel is now empty. The 'Generated Results' panel remains the same. Below the panels are four buttons: 'Clear Generated', 'Add to Selected', 'Remove from Selected', and 'Clear Selected' (which is highlighted with a blue border). At the bottom right of the interface are two more buttons: 'Export Selected' and 'Close'. A status bar at the bottom left of the interface displays the text 'Selected phonewords cleared.'.

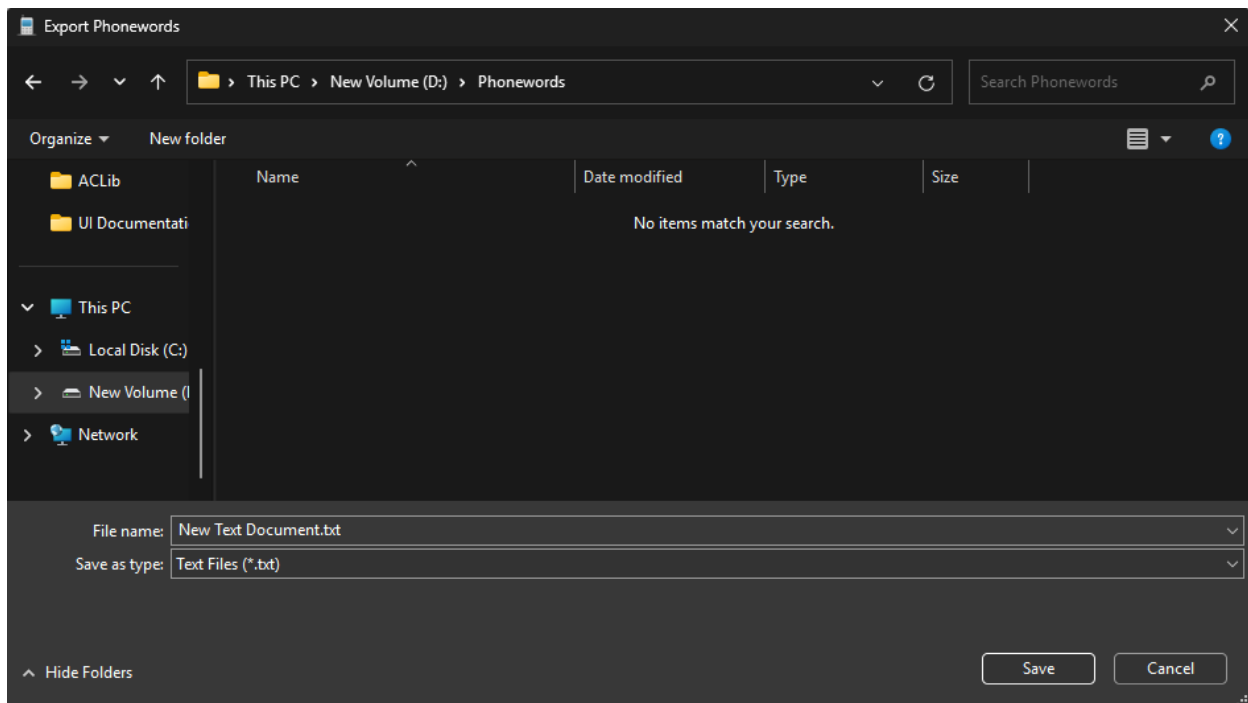
4.4 Exporting Selected Phonewords

4.4.1 Initiate Export

- Ensure there are phonewords in the **Selected List**.
- Click the **Export Button** (`btnExport`).

4.4.2 Save the File

- A **Save File Dialog** appears:
 - Choose where you want to save the file.
 - Enter a file name.
 - Click **Save** to export the selected phonewords.
- A confirmation message appears after the export is successful.



4.5 Performing a Reverse Search

4.5.1 Enter a Word

- Type a word into the **Word Input Textbox** (txtWordInput).
 - Only letters (A-Z) are allowed.
 - The letters are automatically converted to uppercase.

4.5.2 Convert Word to Phone Number

- Click the **Reverse Search Button** (btnReverseSearch).
- The corresponding phone number appears in the **Generated List**.
 - Displayed in the format: **WORD => NUMBER**.

The screenshot shows a web application interface for reverse searching. At the top, there is a label "Enter Word for Reverse Search:" followed by a text input field containing the word "PHONE". To the right of the input field is a button labeled "Reverse Search". Below this, there is a label "Add Custom Words To The Phoneword Dictionary (One per line):" followed by a large, empty text area with a vertical scrollbar. At the bottom, there are two main sections. The left section is titled "Generated Results" and contains a "Search:" label followed by a small text input field. Below this, there is a large text area displaying the result "PHONE => 74663". The right section is titled "Selected Results" and contains a large, empty text area.

4.6 Using Custom Words in Generation

4.6.1 Enter Custom Words

- Type custom words into the **Custom Words Textbox** (`txtCustomWords`), one per line.
 - Only letters are allowed.
 - The text converts to uppercase automatically.

Add Custom Words To The Phoneword Dictionary (One per line):

PHOWN

4.6.2 Generate Phonewords with Custom Words

- After entering custom words, follow the steps in **4.1 Generating Phonewords**.
- The application includes your custom words in the generation process, potentially yielding more personalized results.

Phone Number: 74696

Generate Phonewords

Cancel

Enter Word for Reverse Search:

Reverse Search

Add Custom Words To The Phoneword Dictionary (One per line):

PHOWN

Generated Results

Search:

PHOWN

SHOWN

PI-OWN

SH-OWN

PIN-YO

RHO-YO

RIM-YO

SIN-YO

PG-OWN

PH-OWN

QG-OWN

Selected Results

4.7 Exiting the Application

- Click the **Close Button** (btnClose) to exit the application.
 - Make sure to export any selected phonewords you want to keep before closing.
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5. Design Choices

- **Simple Layout:** All main features are available on one screen for easy access.
 - **Clear Labels:** Each control has a clear label to explain its purpose.
 - **Input Restrictions:** Textboxes accept only valid characters (numbers or letters), reducing errors.
 - **Automatic Formatting:** Text inputs convert to uppercase where appropriate.
 - **User Feedback:** Progress bars, status messages, and tooltips keep you informed.
 - **Easy Navigation:** Grouped controls and intuitive placement make the app user-friendly.
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6. Conclusion

The Phoneword Generator's interface is designed to be straightforward and user-friendly. By following the steps in this guide, you can easily convert phone numbers to words, find the numeric equivalent of words, manage your favorite phonewords, and export them for future use.